C-DAC MUMBAI

Object Oriented Programming using C++

Assigment-1

Q1. Create a class called Student with the following private data members:

1. name (string):to store the name of the student.
2. rollNumber(int):to store the rollnumber of the student.
3. marks(float): to store the marks obtained by the student.
4. grade(char):to store the grade calculated based on the marks.

Implement getter and setter member functions for each data member

Create a member function calculateGrade() that calculates the grade based on the following grading scale:

A: 90-100

B: 80-89

C: 70-79

D: 60-69

F: Below 60

Implement a menu-driven program in the main() function with the following options:

1. Accept Information
2. Display information
3. Calculate Grade
4. Exit the program.

Q2.Create a C++ program for a simple banking system.You need to implement a class called

1. BankAccount with the following data members:
2. AccountNumber(int): The account number of the bank account.
3. accountHolderName(string): The name of the account holder.
4. balance (double):The current balance in the account.

The BankAccount class should have the following member functions:

1. Getter and Setter Methods:

2. Deposit method: A method that allows the user to deposit money into the account. It should take an amount as a parameter and update the balance accordingly.

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3. withdraw method: A method that allows the user to withdraw money from the account. It should take an amount as a parameter and update the balance. Make sure to check if there is sufficient balance before allowing the withdrawal.

4. displayAccountDetails method: A method that displays the account details (account number, account holder name, and balance).

Now, create a menu-driven program in the `main` function that allows the user to perform the following operations:

1. Deposit money into an existing account.
2. Withdraw money from an existing account.
3. Display the account details.
4. Exit the program.

Q3. Imagine you are tasked with creating a program to simulate a toll booth. The toll booth keeps track of the number of vehicles that have passed through it and the total amount of money collected. You need to implement a class TollBooth with appropriate data members and member functions to accomplish this.

Here are the details for the TollBooth class:

1. **DataMembers:**

- totalVehicles: An integer to keep track of the total number of vehicles that have passed through the toll booth.
- totalRevenue: A double to store the total revenue collected from tolls.

2. **Member Functions:**

1. voidreset():Resets both the totalVehicles and totalRevenue to zero.
2. void vehiclePayingToll(int vehicleType, double tollAmount): Accepts an integer vehicleType and a double tollAmount. The vehicleType represents the type of car (1 for standard car, 2 for truck, 3 for bus). The function should increment totalVehicles by 1 and add tollAmount to totalRevenue based on the car type.
3. int getTotalVehicles() : A getter method that returns the total number of vehicles passed through the booth.
4. Double getTotalRevenue(): A getter method that returns the total revenue collected.

3. **Menu-DrivenProgram:**

Write a menu-driven program in main() that allows the user to interact with the TollBooth class:

- Display a menu with the following options:

1. Add a standard car and collect toll

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2. Add a truck and collect toll

3. Add a bus and collect toll

4. Display total cars passed

5. Display total revenue collected

6. Reset booth statistics

7. Exit

- Implement the logic for each menu option using appropriate member functions of the TollBooth class.

- Continue displaying the menu until the user chooses to exit.

- Define a fixed toll amount for each type of car (e.g., RS.180 for standard cars, Rs.300 for trucks, Rs. 350 for buses).

Q4. You are tasked with creating an Employee Payroll Management System in C++. Your program should allow the user to perform the following tasks through a menu-driven interface:

1. Add a new employee:

- Create a class Employee with the following private data members:
- int empID (EmployeeID)
- string empName (EmployeeName)
- double empSalary(Employee Salary)
- Include appropriate getter and setter methods for these data members.
- Ensure that the Employee ID is unique for each employee.

2. Calculate the gross salary for an employee:

- Create a member function calculate GrossSalary in the Employee class.
- The gross salary should be calculated based on the following rules:
- If the employee's salary is less than or equal to 5000, add a 10% bonus.
- If the employee's salary is greater than 5000 but less than or equal to 10000, add a 15% bonus.
- If the employee's salary is greater than10000,add a 20% bonus.
- Display the gross salary for the chosen employee.

3. Display the employee details:

- Create a member function displayEmployeeDetails in the Employee class to display all the details of an employee (ID, Name, Salary, and Gross Salary).

4. Update employee information:

- Allow the user to update the employee's name and salary using setter methods.

5. Exit the program.

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